

Answer Key & Teacher Notes

Session 1 – Simplifying to Show Equivalence • Session 2 – Testing and Justifying Equivalence

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Simplifying to Show Equivalence

TALK PROMPTS

- What has to be true about two terms before you are allowed to combine them?
- In $9r + 7 + 6r + 8 = 15r + 15$, both parts end in 15. Do the two 15s mean the same thing? Explain.

PARTNER TASK

Sorting Terms and Same or Not. One partner sorts the like terms in $4x + 12 + 3x + 5$ and simplifies ($7x + 17$); the other checks by substituting $x = 2$ into both forms. Switch roles with $5n + 5n$ compared with $10n$ and with n^2 .

WATCH FOR

- Combining a constant with a variable term ($12x + 3 + 4x$ as $19x$).
- Treating $5n + 5n$ as n^2 or $10n^2$ instead of $10n$.
- Forgetting that a lone x has a coefficient of 1 ($6 + 2k + k$ as $2k + 6$).

SESSION 2 • TEACHER NOTES

Testing and Justifying Equivalence

TALK PROMPTS

- If two expressions match for one test value, does that prove they are equivalent? Why?
- You tested two values for the ticket formulas, $6t + 5 + 2t$ and $8t + 5$. What else could you do to be sure they always agree?

PARTNER TASK

One Test Is Enough and Properties. One partner tests $3(k + 2)$ and $3k + 2$ with $k = 4$ (18 vs. 14); the other names the property in $2 + (m + 11) = (2 + m) + 11$ and tests $m = 6$ (19). Switch and justify $6t + 5 + 2t = 8t + 5$ with properties.

WATCH FOR

- Deciding two expressions are equivalent after one matching test value.
- Distributing to only the first term ($3(k + 2)$ as $3k + 2$).
- Adding a constant to a coefficient ($6 + 5j$ as $11j$).

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.EE.A.3	1	2	1	1	2	2	3	2	3	1	1
S1 • Form B	6.EE.A.3	1	2	1	1	2	2	3	2	3	1	1
S2 • Form A	6.EE.A.4	1	2	1	1	2	2	2	3	3	1	1
S2 • Form B	6.EE.A.4	1	2	1	1	2	2	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

like terms: Terms with the same **variable** part.

combine: Add or subtract like terms into **one** term.

equivalent expressions: Expressions that have the same **value** for every number.

- Like terms have the same **variable** part; $4x$ and 4 are not like terms.
- Combine like terms by adding their **coefficients**: $4x + 9x = 13x$.
- Two expressions are equivalent if they simplify to the **same** expression.

Example: The like terms with b are $9b$ and $3b$. Combine them: $9b + 3b = 12b$. Combine the constants: $4 + 1 = 5$. Answer: $9b + 4 + 3b + 1 = 12b + 5$

YOU TRY

- a. $8a$ b. $5x + 3$
- Yes. $2n + 3n = 5n$, so $2n + 7 + 3n = 5n + 7$.

EXIT TICKET

- a. $13x$ b. $5y + 5$
- $7k + 6$
- A** ($9n + 2$)

PRACTICE FORM A

- a. $8x + 8$ b. $7m + 4$ c. $7x + 17$ d. $15r + 15$ e. $10n$ f. $3k + 6$
- a. $3p$ and $2p$ b. $3p + 2p = 5p$, so $5p + 10$ c. Yes; Ana's expression simplifies to Luis's, so they are equivalent. d. $3p$ has the variable p and 10 does not, so they are not like terms.
- $12m + 18$
- $11h + 15$
- Rashid is wrong. $4t$ and 16 are not like terms, so $4t + 16$ is already simplified. Test $t = 2$: $4 \times 2 + 16 = 24$, but $20 \times 2 = 40$.
- A: $5a + 10$; B: $5a + 10$; C: $a + 15$. A and B are equivalent.
- a. $7p + 3$ b. Yes, both are $7p + 3$. c. Both equal $7 \times 4 + 3 = 31$. Equivalent expressions give the same value.
- No. $4y$ and 5 are not like terms, so $4y + 5$ cannot become $9y$. For $y = 2$: $4 \times 2 + 5 = 13$, but $9 \times 2 = 18$.
- Only $12x$ and $4x$ are like terms; the 3 has no x . $12x + 4x = 16x$, so the answer is $16x + 3$.
- B** ($8n + 4$)
- C** ($5m$ and 5)

Self-check word TERMS

PRACTICE FORM B

- a. $8y + 9$ b. $7k + 5$ c. $7x + 13$ d. $12r + 15$ e. $8n$ f. $4w + 9$
- a. $4t$ and $3t$ b. $4t + 3t = 7t$, so $7t + 12$ c. Yes; Maya's expression simplifies to Owen's, so they are equivalent. d. $4t$ has the variable t and 12 does not, so they are not like terms.
- $14m + 16$
- $11h + 17$
- Lucia is wrong. $3t$ and 18 are not like terms, so $3t + 18$ is already simplified. Test $t = 2$: $3 \times 2 + 18 = 24$, but $21 \times 2 = 42$.
- A: $6b + 7$; B: $6b + 7$; C: $b + 13$. A and B are equivalent.
- a. $9p + 5$ b. Yes, both are $9p + 5$. c. Both equal $9 \times 8 + 5 = 77$. Equivalent expressions give the same value.
- No. $3y$ and 6 are not like terms, so $3y + 6$ cannot become $9y$. For $y = 2$: $3 \times 2 + 6 = 12$, but $9 \times 2 = 18$.
- Only $10x$ and $6x$ are like terms; the 5 has no x . $10x + 6x = 16x$, so the answer is $16x + 5$.
- A** ($8a$)
- A** (like terms)

Self-check word MATCH

GUIDED NOTES

equivalent expressions: Expressions with the same value for **every** value of the variable.

counterexample: One test value that shows a claim is **wrong**.

Distributive Property: Multiply the number outside by **each** term inside the parentheses.

Commutative Property: You can change the **order** when you add or multiply.

- To test, substitute the **same** value into both expressions.
- One value that gives different results proves they are **not** equivalent.
- Matching values do not prove it; **justify** with properties.

Example: Try $x = 4$: $2 \times 4 + 8 = 16$ and $4 \times 4 = 16$. One match is not proof. Try $x = 1$: $2 \times 1 + 8 = 10$ and $4 \times 1 = 4$. $10 \neq 4$, so $x = 1$ is a **counterexample**. Answer: They are **not** equivalent.

YOU TRY

1. a. 8 b. 7
2. $m = 1$: 5 and 5; $m = 2$: 10 and 6. Not equivalent ($m = 2$ is a counterexample).

EXIT TICKET

1. a. 12 b. 9
2. $m = 1$: 6 and 6; $m = 4$: 15 and 24. Not equivalent.
3. **C** (Distributive)

PRACTICE FORM A

1. a. 32 b. 32 c. 25. $2(y + 7)$ and $2y + 14$ might be equivalent (the Distributive Property shows they are).
2. a. A: $6 \times 3 + 5 + 2 \times 3 = 29$; B: $8 \times 3 + 5 = 29$ b. A: 85; B: 85 c. Yes. Reorder and combine: $6t + 2t + 5 = 8t + 5$. d. Commutative Property of Addition
3. $3(4 + 2) = 18$; $3 \times 4 + 2 = 14$. $18 \neq 14$, so not equivalent.
4. a. Associative Property of Addition b. Both equal 19.
5. $6 + 5 \times 2 = 16$, but $11 \times 2 = 22$. $16 \neq 22$, so they are not equivalent; 6 and 5j are not like terms.
6. $v = 1$: 15 and 15; $v = 7$: 69 and 69. Yes; the Distributive Property gives $3(3v + 2) = 9v + 6$.
7. $12 \times 5 - 4 \times 5 = 40$, but $8 = 8$. Not equivalent. $12m - 4m = 8m$; the m stays.
8. a. 18, 18, 10 b. $3(k + 4)$ and $3k + 12$ c. Distributive Property: $3(k + 4) = 3 \times k + 3 \times 4 = 3k + 12$.
9. One matching value does not prove equivalence. For $p = 2$: $4 \times 2 + 6 = 14$, but $10 \times 2 = 20$, so they are not equivalent.
10. **B** ($2x + 10$)
11. **B** (not equivalent)

Self-check word PROPERTY

PRACTICE FORM B

1. a. 36 b. 36 c. 27. $4(y + 3)$ and $4y + 12$ might be equivalent (the Distributive Property shows they are).
2. a. A: $4 \times 2 + 7 + 5 \times 2 = 25$; B: $9 \times 2 + 7 = 25$ b. A: 61; B: 61 c. Yes. Reorder and combine: $4b + 5b + 7 = 9b + 7$. d. Commutative Property of Addition
3. $5(3 + 1) = 20$; $5 \times 3 + 1 = 16$. $20 \neq 16$, so not equivalent.
4. a. Associative Property of Addition b. Both equal 17.
5. $7 + 2 \times 3 = 13$, but $9 \times 3 = 27$. $13 \neq 27$, so they are not equivalent; 7 and 2j are not like terms.
6. $v = 1$: 12 and 12; $v = 5$: 44 and 44. Yes; the Distributive Property gives $4(2v + 1) = 8v + 4$.
7. $15 \times 4 - 6 \times 4 = 36$, but $9 = 9$. Not equivalent. $15n - 6n = 9n$; the n stays.
8. a. 16, 16, 11 b. $2(k + 5)$ and $2k + 10$ c. Distributive Property: $2(k + 5) = 2 \times k + 2 \times 5 = 2k + 10$.
9. One matching value does not prove equivalence. For $p = 3$: $4 \times 3 + 6 = 18$, but $10 \times 3 = 30$, so they are not equivalent.
10. **A** (not enough to prove equivalence)
11. **B** ($5a + 10$)

Self-check word CONSTANT

SHEET 1 · SIMPLIFYING TO SHOW EQUIVALENCE

1. a. $9a$ b. $6y$ c. $6x + 2$ d. $3n + 11$ e. $5b + 7$ f. $7p + 9q$
g. $5m + 11$ h. $11z$
2. $2x + (x + 4) + (x + 1) = 4x + 5$ cm
3. No. $3w + 3w = 6w$, not $6w^2$ (for $w = 2$: 12 vs. 24).
4. $7x + 3$ (combine $5x + 2x = 7x$; the 3 has no x)
5. She combined the constant 4 with the x -terms ($7 + 4 - 2 = 9$). Only like terms combine: $7x - 2x = 5x$, so the answer is $5x + 4$.

SHEET 2 · TESTING AND JUSTIFYING EQUIVALENCE

1. a. 27 b. 27 c. 19. $3(x + 4)$ and $3x + 12$ might be equivalent (the distributive property shows they are).
2. a. E b. N c. N d. E
3. Sample: $n = 3$ gives $n^2 = 9$ but $2n = 6$, so they are not equivalent.
4. $x = 4$: $3x = 12$ but $x + 6 = 10$, so they are not equivalent (one test value that gives different results is enough).
5. Distributive
6. **B** ($7p + 21$)
7. **D** ($4(y + 8)$)